COMPANION TO THE FIELD HANDBOOK

Sensor Reference

Every sensor in the system — what it measures, how its signal reaches the processor, and what the engine actually does with the number. Including the ones that belong to the panel rather than the ECU, and the ones we would add.

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00 How to read an entry

Every sensor below has the same five rows, because those are the five things you need to know before you can either build its front-end or write its firmware:

- **Measures** — the physical quantity and its useful range.
- **Signal** — the electrical form it arrives in, its ECU pins, and its wire class from the Kirloskar diagram's own colour legend.
- **Acquisition** — the front-end circuit, which MCU peripheral reads it, and how often.
- **Used for** — what the engine does with the number. This is the row that matters and the one most sensor documentation leaves out.
- **Fault detection** — how firmware knows the reading is lying.

Sampling rates are proposals, not measurements — they follow from each signal's own bandwidth and from what the control loop using it needs. Marked **inferred** where that is the case.

> **A NOTE ON SAMPLE RATE**
>
> The instinct is to sample everything fast. Resist it. Coolant temperature has a thermal time constant measured in tens of seconds, so sampling it at 1 kHz buys nothing and costs ADC bandwidth that rail pressure genuinely needs. Rate should follow the physics, and the physics here spans four orders of magnitude — from crank edges every 667 µs to a fuel level that moves over hours.

01 Position & speed

Everything angle-related derives from these two. Get them wrong and nothing else in the system can be right, because injection timing is meaningless without knowing where the crankshaft is.

Crankshaft speed

pins 52 / 74 / 30

ON ECU

MEASURES	Crank angular position and speed, from a toothed wheel. Nominal running speed **1500 rpm**; must also resolve cranking speed, roughly 100–250 rpm.
SIGNAL	**Variable reluctance**, differential. 🔵 `Frequency I/P High` on pin 52, ⚫ `Frequency I/P Low` on pin 74, ⚫ `sensor ground` on pin 30. Twisted pair. Output is AC whose amplitude scales with speed — near-zero while cranking, several volts at idle.
ACQUISITION	Zero-cross adaptive-threshold conditioner producing one clean digital edge per tooth, into a FlexTimer input capture channel. Every edge is timestamped; at 1500 rpm with a 60-tooth wheel that is an edge every **667 µs**. The missing-tooth gap gives the angular reference.
USED FOR	**Speed governing** — the primary feedback for the metering unit loop. **Injection timing** — every injection event is scheduled against crank angle derived from here. **Overspeed shutdown** — the ECU's own first line of defence. **Crank sequencing** — the starter is disengaged the moment speed crosses the catch threshold. **Reported over CAN** in EEC1, PGN 61444.
FAULT DETECTION	Missing edges, implausible tooth intervals, or loss of the sync gap. Loss of crank signal on a running engine is a shutdown, not an alarm — the ECU can no longer time injection.

Camshaft position `pins 45 / 46 / 44` ON ECU

MEASURES	Which half of the four-stroke cycle the crank angle refers to. One pulse per two crank revolutions.
SIGNAL	**Hall effect**, already a clean digital square wave — which is why it needs no conditioning IC and has no Low pin. 🔴 `Switch Battery 5 V` on pin 45, 🔵 `Frequency I/P` on pin 46, ⚫ `sensor ground` on pin 44.
ACQUISITION	Pull-up to the protected sensor rail, straight into a second FlexTimer capture channel. One event per 80 ms at running speed.
USED FOR	**Cylinder phase**, and only that. The crank sensor says the engine is at 15° before top dead centre; the cam sensor says whether that is cylinder 1's compression stroke or its exhaust stroke. Without it you can measure speed perfectly and still inject into the wrong cylinder.
FAULT DETECTION	Absent pulse, or a pulse at the wrong crank angle relative to the sync gap. Loss after sync is survivable for a while — the engine can run on the last known phase — but it must not be allowed to re-sync blind.

02 Fuel system

Rail pressure `pins 32 / 35 / 08` ON ECU

MEASURES	Common-rail fuel pressure. Typically **250–1800 bar** on an engine of this class — **inferred**, the exact range needs the sensor part number.
SIGNAL	Ratiometric **0.5–4.5 V**. 🔴 `Switch Battery 5 V` on pin 32, 🟣 `Analog I/P` on pin 35, ⚫ `sensor ground` on pin 08 — a **dedicated** ground, not spliced with any other sensor. That is deliberate: this is the most accuracy-critical analog channel on the engine, and a shared return would inject other sensors' currents into its reference.
ACQUISITION	Ratiometric front-end into the ADC. Sampled fast — **1 kHz** or better, **inferred** — because it is the feedback for a closed loop that has to reject the pressure ripple every injection event causes.
USED FOR	**Closed-loop rail pressure control**: the error between measured and target pressure drives the fuel metering unit's PWM duty. Also sets the **injection quantity calculation** — the same pulse width delivers more fuel at higher rail pressure, so quantity is always a function of both. **Over-pressure protection** is a hard shutdown.
FAULT DETECTION	Out-of-range reading, or pressure that fails to follow a metering unit command — which distinguishes a dead sensor from a stuck metering unit. Loss of rail pressure feedback means injection quantity becomes unknown; treat as shutdown.

Fuel metering unit current `pin 88, sensed on-board` ON ECU

MEASURES	Current through the metering unit solenoid — the actuator's own feedback, not an external sensor.
SIGNAL	Shunt and amplifier on the ECU's low-side driver. The metering unit itself is only 2-pin: 🔴 `battery` via the 20 A fuse, and ⚫ `PWM low side` to pin 88.
ACQUISITION	ADC channel synchronised to the PWM cycle, so current is sampled at a consistent point in the switching period rather than at random phase.
USED FOR	**Inner loop of the rail pressure cascade.** Rail pressure sets a target current; current control sets PWM duty. Two loops because solenoid force follows current, and current follows duty only loosely once coil temperature and battery voltage move. Also gives **open and short circuit detection** on the actuator.
FAULT DETECTION	Zero current with duty commanded means an open coil or blown fuse. Full current at zero duty means a shorted driver — which is dangerous, since it drives the rail to maximum pressure.

Injector current, ×3 `pins 73 / 07 / 29, sensed on-board` ON ECU

MEASURES	Current through each injector solenoid during its firing event. Peak **12–24 A**, hold **8–13 A**.
SIGNAL	Sense resistor in each low-side leg, read by the driver IC's own current comparators.
ACQUISITION	The MC33816 regulates current in hardware against programmed thresholds — the MCU sets the profile, the driver enforces it microsecond by microsecond. Far too fast for firmware to close.
USED FOR	**Peak-and-hold profile enforcement.** A high current opens the needle, then a lower current holds it, because holding at peak would cook the solenoid. Also **injector diagnosis**: a coil that fails open or short shows up immediately in the current trace.
FAULT DETECTION	Failure to reach peak within the expected time means an open coil or a collapsed boost rail. The handbook's injector chart shows why that timing is diagnostic — the rise time is set by the rail voltage and the coil, so a slow rise means one of them is wrong.

03 Air path

Boost pressure `pins 11 / 41 / 34` ON ECU

MEASURES	Absolute pressure in the intake manifold, downstream of the turbocharger and aftercooler. Typically **1–3 bar absolute** on a turbocharged engine of this size — **inferred**.
SIGNAL	Ratiometric 0.5–4.5 V. One half of a **four-pin combination sensor** shared with manifold air temperature: 🔴 `Switch Battery 5 V` on pin 11, 🟣 `Analog I/P Pressure` on pin 41, ⚫ `sensor ground` on pin 34 shared with both legs and with the coolant sensor.
ACQUISITION	Ratiometric front-end into the ADC. **100 Hz** is ample — manifold pressure changes with turbo speed, which has real inertia. **Inferred.**
USED FOR	**Air mass estimation**, with air temperature, via the ideal gas law — and air mass is what sets the **smoke-limit fuelling cap**. A diesel that injects more fuel than its air charge can burn produces visible smoke and heat rather than torque, so the fuelling limit tracks boost. Also **EGR control**, since EGR displaces fresh air, and **turbo overspeed protection**.
FAULT DETECTION	Range check, plus plausibility against barometric pressure at rest — with the engine stopped, manifold pressure must read close to ambient. A sensor that fails high would silently allow over-fuelling, so this one defaults conservative on fault.

Manifold air temperature `pin 79`

TOPOLOGY OPEN — U2

MEASURES	Intake charge temperature after the aftercooler. Roughly **−20 to +100 °C**.
SIGNAL	`Analog I/P Temperature` on pin 79, sharing the combination sensor's supply and ground. **Whether it is an NTC or a conditioned ratiometric output is unresolved.** The wire colour class cannot distinguish them — the coolant NTC uses the same magenta class. Industry convention for combination pressure-and-temperature sensors is an NTC element, so that is the working assumption.
ACQUISITION	Pull-up divider if NTC, ratiometric front-end if not. The board carries a **DNP pull-up footprint** so either works without a respin — the cheapest possible insurance against a question we cannot currently answer.
USED FOR	**Air density correction.** Boost pressure alone does not give air mass; the same pressure at 20 °C and at 80 °C is meaningfully different charge. Feeds the same smoke-limit calculation. Also **intercooler diagnostics** — a rising delta over ambient means the aftercooler is fouling.
FAULT DETECTION	Range check, and plausibility against coolant temperature at cold start: on a machine that has stood overnight, intake air, coolant and ambient should all agree within a few degrees. That check is free and catches a surprising number of wiring errors.

EGR valve position

pins 15 / 37 / 36 — ON ECU

MEASURES	Actual position of the EGR valve, 0–100% travel.
SIGNAL	Ratiometric potentiometer. 🔴 `Switch Battery 5 V` on pin 15, 🟣 `Analog I/P` wiper on pin 37, ⚫ `sensor ground` on pin 36. Shares its five-pin connector with the actuator's own 🟣 `EGR High` and ⚫ `EGR Low` drive pins.
ACQUISITION	Ratiometric front-end into the ADC, **100 Hz**, matched to the valve's mechanical response. **Inferred.**
USED FOR	**Closed-loop EGR position control** — the H-bridge drives until measured position matches demand. EGR rate is one of the two things this engine uses to meet its NOx limit, the other being injection timing, which is exactly why the certification question in the handbook bites so hard. Also **stuck-valve detection**.
FAULT DETECTION	Position failing to follow demand within a timeout means a stuck valve, a failed motor, or a broken position sensor — distinguishable by whether current is flowing. A stuck-open EGR valve at load is a derate condition.

Intake throttle position

6-pin connector, routing unknown — UNRESOLVED — U8

MEASURES	Intake throttle plate position. On a CPCB IV+ diesel the throttle is not a power control — it creates intake depression to drive EGR flow when the pressure differential is otherwise insufficient.
SIGNAL	**Six pins**, which is two more than a motor pair plus a three-wire position sensor accounts for. The extra two are unexplained. It does not appear on any ECU pin traced in the wiring diagram — it may belong to a variant, or route somewhere we have not found.
ACQUISITION	Not designed. Blocked until the connector is traced on the engine.
USED FOR	If present: **EGR flow authority at low load**, and **a clean shutdown** — closing the throttle starves the engine of air and stops it smoothly.
FAULT DETECTION	Not designed.

04 Engine health

These are the sensors that stop the engine before it destroys itself. They earn their place in an ECU that never optimises anything.

Coolant temperature `pin 33` ON ECU

MEASURES	Engine coolant temperature, **−40 to +130 °C**. Normal running is roughly 80–95 °C.
SIGNAL	**NTC thermistor, two wires only** — no excitation pin, unlike the ratiometric sensors. 🟣 `Analog I/P` on pin 33; its ⚫ `sensor ground` does not get its own ECU pin but splices into the boost sensor's return at pin 34.
ACQUISITION	Pull-up divider into the ADC, linearised in firmware against the sensor's resistance-temperature table. **2 Hz is plenty** — an engine block's thermal time constant is tens of seconds. The pull-up is chosen to put maximum sensitivity around 85 °C, where the decisions are.
USED FOR	**Cold-start fuelling enrichment** — a cold diesel needs more fuel and earlier injection to light off. **Overheat derate then shutdown. EGR inhibit when cold**, since EGR on a cold engine fouls the intake. **Reported over CAN** in ET1, PGN 65262.
FAULT DETECTION	Open circuit reads full pull-up voltage and short reads zero — both outside any physical temperature, so both are unambiguous. Critically, **an open circuit must not read as "very cold"**, which would trigger enrichment on a hot engine. The independent coolant switch below exists precisely because this channel can fail.

Coolant temperature switch `pin 43, return on 01 / 02` ON ECU

MEASURES	A single over-temperature threshold. Not a measurement — a trip.
SIGNAL	🔵 `Digital I/P` on pin 43, switch-to-return. Its return leg is one of the `G_G_B` pins.
ACQUISITION	Pull-up and RC debounce into a GPIO, polled at **10 ms**.
USED FOR	**A second, independent path to overheat shutdown.** This is the same design instinct as the overspeed discussion in the handbook: the analog coolant channel and this switch fail in different ways, so requiring only one of them to trip means a single sensor failure cannot leave the engine cooking. It is cheap redundancy on the failure that quietly destroys engines.
FAULT DETECTION	Wired normally-closed where possible, so an open harness reads as tripped rather than as healthy.

Lubricating oil pressure `pins 39 / 80` ON ECU

MEASURES	Engine oil gallery pressure, typically **0–10 bar**. Normal running above roughly 2 bar; idle above roughly 1 bar. **Inferred.**
SIGNAL	Ratiometric 0.5–4.5 V, three pins: 🔴 `Switch Battery 5 V` on pin 39, 🟣 `Analog I/P` on pin 80, and a ⚫ `sensor ground` that splices with the EGR sensor's into pin 36.
ACQUISITION	Ratiometric front-end into the ADC at **10 Hz**. **Inferred** — oil pressure responds within a revolution, but no decision needs faster.
USED FOR	**Low oil pressure shutdown**, the single most important protection on any engine — bearing damage from oil starvation happens in seconds. The threshold is **speed-dependent**: pressure that is normal at idle is a fault at rated speed, so the limit is a function of the crank sensor's reading. Also **start inhibit** until pressure builds, and reported over CAN in EFL/P1, PGN 65263.
FAULT DETECTION	Range check. The dangerous failure is reading high when pressure is actually low, so a plausibility check against engine state helps: with the engine stopped, oil pressure must read near zero. A sensor that reads 3 bar on a stopped engine is lying.

05 Electrical

Battery voltage, ×2 pins 04 / 06 ON ECU

MEASURES	Voltage on two separately fused battery rails, **6–40 V** covering cranking sag through clamped load dump. Nominal 12 V, charging 13.5–14.4 V.
SIGNAL	Two 🔴 `DC positive` feeds, `V_V_BAT_1R` and `V_V_BAT_2R`, each behind its own 10 A fuse.
ACQUISITION	**120 kΩ / 10 kΩ** divider into the ADC — no active conditioning, this is a slow rail rather than a delicate signal. **10 Hz.** The divider value was corrected from an inherited 75 kΩ / 10 kΩ that would have put 4.69 V on a 3.3 V pin.
USED FOR	**Low and high battery alarms. Charging system diagnosis** — voltage that never rises above resting means the alternator is not charging. **Injector timing compensation**, since boost converter recharge time depends on supply voltage. And because the two rails are separately fused, **comparing them infers a blown fuse** without a dedicated fuse-status circuit. Reported over CAN in VEP1, PGN 65271.
FAULT DETECTION	The two channels check each other: they should track within a volt. A persistent divergence means a fuse, a corroded connection, or a failed divider.

Ignition / run request `pin 71` ON ECU

MEASURES	Whether the panel is commanding the engine to run.
SIGNAL	🔴 `Switched battery`, active high. Downstream of the GCU's 70 A ignition relay.
ACQUISITION	Zener-clamped bias network into a GPIO — not a divider, for the reason given in the handbook. Polled at **10 ms**.
USED FOR	**Run enable and orderly shutdown.** Losing this input is the normal stop request: the ECU stops injection, closes the throttle if fitted, and reports the stop over CAN. It is also, per the handbook's safety section, the **signalling half of a path whose other half is physical** — the same relay that drops this input is believed to remove power from the fuel system entirely, which is what makes a hung ECU survivable.
FAULT DETECTION	Not detectable as an open circuit — an open wire and a genuine stop request are electrically identical. That is an accepted limitation of matching the OEM harness, and it is recorded in the FMEA rather than engineered around, because changing the sense would break compatibility.

06 Operator inputs

Not sensors in the strict sense, but they arrive on the same connector and firmware treats them identically — debounced digital state with a defined meaning.

INPUT	PIN	SENSE	WHAT IT DOES	STATUS
Audio abort	20	🔴 `active high`	Silences the buzzer without clearing the underlying fault. The fault stays active, the annunciator stays lit, and only the noise stops — so an operator can think without losing the alarm.	Confirmed
Override SS	24	🔴 `active high`	Function genuinely unconfirmed. Labelled `OVERSIDE SWITCH` at the ECU and `-S7 Override SS SW` at the panel — **one switch, not two**, despite earlier documents listing an overspeed switch and an override switch separately.	Open
Droop select	23	🔵 `switch to return`	Selects droop versus isochronous governing. In droop, speed falls slightly as load rises, which lets multiple sets share load without fighting. Isochronous holds speed exactly, which is right for a set running alone.	Confirmed

> **DO NOT WIRE CONTROL LOGIC TO PIN 24 YET**
>
> The override switch's actual function is a guess. Read its state, log it, report it over CAN — but do not let it change what the engine does until somebody has confirmed what the panel intends by it. A switch whose meaning you have assumed is worse than a switch you ignore.

WHAT THE CONTROLLER NOW DEMANDS FROM US — CONFIRMED 18 SEP 2026

The fitted **DSE4522**'s own configuration takes **oil pressure**, **coolant temperature**, **coolant level** and **engine speed** from the ECU over CAN rather than from its own analogue inputs, on source addresses 234 (engine) and 44 (instrumentation). Its analogue inputs A and B are configured *Not Used* for exactly that reason.

Two consequences for this design. Those four values are no longer optional instrumentation — the controller's protections act on them, at 0.99 bar oil pressure and 200 °C coolant. And **ECU Data Fail is configured to shut the set down**, so our CAN transmission is itself a watchdog held by another device. Engine speed is the uncomfortable one: the controller's overspeed trip reads the number we send, which is why handbook §04 now calls a standalone trip module mandatory.

07 Panel-side sensors

These exist in the machine and appear in the wiring diagram, but they do not reach the ECU. They belong to the panel controller — a Deep Sea **DSE4522 MKII AMF**, identified 18 Sep 2026; earlier revisions of this page said KG640C, which is what the OEM wiring diagram shows but not what is fitted. They are documented here because earlier design passes built ECU front-ends for two of them, and because knowing what the panel already measures tells you what the ECU does not need to.

SENSOR	GOES TO	SIGNAL	USED FOR
Fuel level	Panel via XC pin 22	Resistive float sender, 2-pin	Tank level display and low-fuel alarm. **Not an ECU signal** — earlier documents wrongly gave it an ECU front-end.
Water in fuel	Panel via XC pin 16/1	Conductivity probe, 3-pin with built-in electronics	Water-in-fuel warning. Water in a common-rail system destroys injectors, so this is a genuine alarm. **Not an ECU signal.**
Catalyst temperature	Unresolved — U9	2-pin, one sensor only	DOC inlet or outlet temperature. Earlier documents specified three EGT channels; the diagram shows one.

SENSOR	GOES TO	SIGNAL	USED FOR
Alternator voltage & frequency	GCU terminals 27–30	3-phase AC sense, sampled at 5 kHz	Output metering, and — per the handbook's safety section — **the GCU's only engine speed source**. On a 4-pole alternator 1500 rpm is rigidly 50 Hz, which is what makes overspeed detection possible without an ECU.
Load current	GCU terminals 35–42	Current transformers	Load metering, overload protection, load-dependent start/stop.
Charging alternator D+	GCU	Excitation sense	Charge-failure alarm, and a secondary indication the engine is turning.

08 What we would add

The OEM sensor set is the minimum to run the engine and meet emissions. These are the additions worth designing space for now, since a connector position or an unused ADC channel costs almost nothing at design time and cannot be retrofitted cheaply.

Ordered by how much each earns its place.

ADDITION	INTERFACE	WHY IT EARNS A PLACE	COST TO DESIGN IN NOW
Barometric pressure	On-PCB MEMS, I²C	Altitude and weather change air density, and therefore the smoke-limit fuelling cap. Without it the engine either smokes at altitude or is derated everywhere to be safe. Needs no harness at all — barometric pressure is the same inside the enclosure as outside.	One IC, two I²C pins, a vented enclosure port. **Already in the design.**
Magnetic pickup	VR, to a standalone trip module	The independent overspeed path discussed in the handbook. Currently recommended rather than mandatory, because the GCU's alternator-frequency path already bypasses the ECU — but it is the only option that also survives a GCU firmware hang.	A flywheel sensor, a trip module, and panel wiring. No ECU pins at all, which is the point.
Exhaust gas temperature, ×2	Type K thermocouple, CJC amplifier	DOC inlet and outlet. The difference across a catalyst tells you whether it is working; a single sensor tells you much less. Also protects the turbine from over-temperature.	Two amplifier ICs, two ADC channels, four connector positions.
Fuel temperature	NTC, 2-wire	Diesel density varies with temperature, so the same injected volume is a different mass. Matters most for fuel consumption reporting and for hot-restart behaviour, where return fuel has been heated.	One ADC channel, one pull-up, two connector positions.

ADDITION	INTERFACE	WHY IT EARNS A PLACE	COST TO DESIGN IN NOW
Crankcase pressure	Low-range ratiometric	The best single indicator of engine wear available. Rising blow-by past the rings shows here long before anything else notices, which turns an unplanned failure into scheduled maintenance.	One ADC channel and a sensor. Genuinely the highest-value addition for a machine expected to run for years.
Ambient air temperature	NTC, 2-wire	Separates engine-related temperature rise from a hot day. Makes the derate strategy fair, and makes the aftercooler diagnostic in §03 meaningful.	One ADC channel.
Accelerometer	MEMS, SPI or I²C, on-PCB	Vibration signature as a condition-monitoring input — bearing wear, misfire, and mounting problems all have signatures. Speculative for a genset, but nearly free if the board is being laid out anyway.	One IC and a few pins. Firmware effort is the real cost, not hardware.

WHAT FUTURE-PROOFING ACTUALLY MEANS HERE

Not fitting every sensor on the list. It means **reserving the channels and connector positions** so that adding one later is a firmware change and a harness leg, not a board respin. Concretely: leave four ADC channels unrouted but broken out, keep two spare positions on the sensor connector, put the I²C bus on a header even if only the barometric sensor uses it now, and leave the second CAN channel populated.

The MCU has room for all of it — the channel budget below shows the design using well under half of what the part provides. Spending some of that headroom on unpopulated options is the cheapest decision available at this stage.

09 Channel budget

What the sensor set costs in MCU resources, against what the S32K148 provides. The headroom is the argument for designing the additions in now.

RESOURCE	USED NOW	WITH ALL ADDITIONS	S32K148 PROVIDES	HEADROOM
ADC channels	14	19	up to 64	ample
Timer capture	2	2	64 across 8 FlexTimers	ample
Timer output / PWM	8	8	shared with above	ample
Digital in	5	5	abundant GPIO	ample
CAN	2	2	3 FlexCAN	1 spare
I²C	1	1	2 LPI2C	1 spare
SPI	1 (injector driver)	2	3 LPSPI	1 spare

The ADC count is the one to watch, and it is not close to a limit. The caveat from the MCU memo stands: how many of those 64 channels are actually broken out on the 144-pin package needs the reference manual's pin-mux table, which is a Phase 2 item.